

Ψ -Risk-DT: A Neurosymbolic Framework for Adaptive Zero-Day Threat Detection in Digital Twin Ecosystems

Abstract

Digital Twin (DT) ecosystems—including low-power IoT networks, cyber-physical systems, and virtual replicas of industrial assets—are highly vulnerable to zero-day threats, i.e., novel attacks exploiting unknown vulnerabilities. These environments face challenges such as heterogeneous device topologies, evolving data distributions (concept drift), and limited computational resources, all of which undermine traditional Intrusion Detection Systems (IDSs). Existing methods often produce high false positives due to opaque statistical models, or fail to adapt to new threats because of rigid rule-based logic.

To address this, we introduce Ψ -Risk-DT, a neurosymbolic framework that integrates entropy-based anomaly detection with an Associated Random Neural Network (ARNN) and a semantic reasoning layer based on RDF/SPARQL. A key neurosymbolic operator, Ψ , couples neural pattern recognition with dynamic semantic updates, triggering adaptive analysis when entropy spikes indicate potential zero-day behaviour. Additionally, a Modular Semantic Update (MSU) mechanism rewrites only the relevant portions of the knowledge graph, minimizing computational overhead in resource-constrained environments. The architecture is optimized via a hybrid loss function balancing detection accuracy, graph consistency, and semantic alignment.

Evaluated on out-of-distribution variants of established IoT/DT benchmarks (Kitsune, WiseML 2024, Sec4ML 2023), Ψ -Risk-DT detects zero-day threat patterns without relying on known signatures, achieving 85–95% detection rates for unseen attacks. It reduces false positives by 20–40% compared to state-of-the-art baselines, delivers sub-second latency (≈ 247 ms, a 30% improvement over neural-only IDSs), and cuts symbolic processing load by up to 88% through MSU.

Overall, Ψ -Risk-DT provides resilient and explainable protection for DT ecosystems, enabling real-time identification of zero-day threats with low overhead and traceable reasoning pathways. This advances neurosymbolic cybersecurity for IoT and paves the way for adaptive defenses in critical infrastructures.

Keywords

Digital Twins; Zero-day threat detection; Neurosymbolic cybersecurity; Intrusion Detection Systems; Entropy-driven anomaly detection; RDF/SPARQL reasoning; Explainable AI

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1 Introduction

Digital Twin ecosystems (DT), cyberphysical environments in which physical assets are mirrored by their virtual counterparts, are rapidly permeating safety-critical infrastructures such as smart manufacturing, energy grids, and connected healthcare [16]. In these settings, streams of multimodal telemetry link physical sensors to cloud or edge services through low-power wireless networks (e.g., RPL, 6LoWPAN). Despite their potential for predictive maintenance and operational intelligence, DT ecosystems remain acutely exposed to zero-day threats that exploit unknown vulnerabilities, topology dynamics, and distributional drift across heterogeneous devices [4].

Conventional Intrusion Detection Systems (IDSs) are ill-suited to this scenario. They typically suffer from high false positive rates, subsecond latency violations, and poor adaptability to unseen attack patterns under resource constraints [1]. Recent surveys on neurosymbolic AI and DT cybersecurity highlight a fundamental limitation: current IDSs cannot reconcile statistical learning (fast but opaque) with symbolic reasoning (interpretable but rigid) in distributed real-time IoT environments [5, 16].

This paper argues that entropy-aware neurosymbolic reasoning provides a principled pathway to bridge this gap. Entropy quantifies uncertainty and information imbalance, while neurosymbolic coupling enables bidirectional alignment between neural activations and symbolic knowledge graphs. By combining these perspectives, a DT security framework can reason over both the probabilistic dynamics of anomalies and the semantic structure of the environment.

Problem Statement. How can neurosymbolic methods be integrated into Digital-Twin security frameworks to achieve resilient, explainable defense against zero-day attacks in low-power IoT networks?

Objective. We propose Ψ -Risk-DT, a neurosymbolic framework that extends the detection of anomalies based on entropy with an operator Ψ coupling Associated Random Neural Networks (ARNN) and RDF/SPARQL reasoning. Ψ performs injection of neural activations based on entropy with injection of entropy into semantic graphs, aligning statistical evidence with contextual knowledge to maintain stability and explainability [15].

Contributions. A formal operator Ψ for conditional RDF updates, integrating ARNN activations and entropy deviation signals within DT topologies.

- (1) A Modular Semantic Update (MSU) mechanism that partitions DT graphs into local components (nodes, sensors, flows), enabling scalable, coherent updates without full-graph rewrites.

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- (2) A hybrid loss function that jointly optimizes classification accuracy, graph coherence, and semantic consistency, with Lyapunov-style stability guarantees.
- (3) An explainable interface (Ψ -UI) that visualizes hybrid neural-symbolic threat graphs, supporting the interpretability of the analyst and the forensic reasoning.

An extensive evaluation on out-of-distribution IoT/DT datasets (Kitsune, WiseML 2024, Sec4ML 2023), demonstrating a 20-40 reduction in false positives and a -30 latency improvement, validated on RPL-based wireless networks.

The remainder of this paper is organized as follows. Section 2 surveys related work in anomaly detection, neurosymbolic reasoning, and DT security; Section 3 presents the proposed architecture; Section 4 details methodology and theoretical analysis; Section 5 reports empirical validation; Section 6 concludes with future directions.

2 Related Work

This section reviews previous contributions in anomaly detection, neurosymbolic reasoning, and semantic knowledge integration for cybersecurity in Digital-Twin (DT) and Internet-of-Things (IoT) environments. We emphasize the open methodological and scalability gaps that motivate the Ψ -Risk-DT framework.

2.1 Evolution of anomaly detection in IoT / DT systems

Early intrusion detection systems (IDS) relied on static rule matching and statistical profiling, achieving limited generalization in non-stationary traffic [4, 6]. The rise of deep learning improved detection accuracy, but introduced opacity, high computational cost, and susceptibility to distributional drift.

Benchmarks such as Kitsune [9] and more recent datasets (WiseML 2024, Sec4ML 2023) have repeatedly highlighted persistent operational challenges, including elevated false-positive rates and latency that can be incompatible with time-critical wireless deployments.

In addition to widely adopted benchmarks such as Kitsune, earlier efforts have focused on the construction of intrusion detection datasets aimed at characterizing evolving traffic patterns and attack variability, providing a basis for benchmarking IDS behavior under distributional drift [14].

Routing Protocol for Low-Power and Lossy Networks (RPL) has been shown to expose distinctive attack surfaces related to topology manipulation, rank inconsistencies, and control-message abuse, as systematically documented in recent surveys on RPL-based threats [11].

Recent lightweight IDS proposals [1] exploit feature selection and online learning; however, they largely remain black-box and provide limited semantic traceability under resource constraints. In low-power RPL-based networks, recent work has explored Associated Random Neural Networks (ARNN) to model probabilistic attack propagation over constrained topologies, suggesting their suitability for routing-layer security analysis in RPL-based IoT deployments [2].

These limitations are magnified by constrained memory and energy budgets, where global model updates or graph rewrites are infeasible. Consequently, anomaly-detection pipelines still lack the

capacity to balance precision, explainability, and adaptivity under zero-day conditions.

Systematic reviews of IoT anomaly detection indicate that many lightweight and online approaches tend to trade detection accuracy for computational efficiency, while often remaining semantically opaque—an issue that becomes particularly relevant in Digital Twin environments. [1, 16].

To strengthen reproducibility and critical comparison, Table 1 summarizes the existing work against Ψ -Risk-DT, highlighting unique features such as entropy-gated coupling and modular RDF updates.

Recent deep learning approaches report promising detection accuracy for zero-day threats, yet often suffer from unstable false-positive rates and reduced robustness under distributional drift, especially when evaluated on benchmark-driven scenarios [4].

2.2 Advances in Neurosymbolic Artificial Intelligence

Neurosymbolic AI (NeSy-AI) seeks to merge neural adaptability with symbolic interpretability [5, 15].

Neurosymbolic intrusion detection frameworks have recently demonstrated improved explainability by combining deep representations with rule-based reasoning. However, such systems are generally applied to fixed datasets or rely on symbolic layers added after the fact, and do not provide entropy-gated, closed-loop integration with streaming telemetry, as required in Digital Twin ecosystems [5, 12].

Graph-based neurosymbolic frameworks and ontology-enhanced cyber-defense systems [12] demonstrate improved reasoning and verifiability in cyber-physical environments. However, existing implementations are primarily offline or simulation-bound, rarely sustaining real-time performance in distributed IoT or embedded contexts.

Formal reasoning on network protocol semantics, such as NTP or RPL, has been explored [13], but without entropy-modulated feedback or closed-loop integration with neural activations. Similarly, hybrid approaches based on GNN [9] address structural reasoning but omit dynamic entropy modeling, which is critical to capture temporal uncertainty in DT synchronization. The absence of entropy-aware coupling remains a core gap that limits neurosymbolic deployment in resource-constrained wireless infrastructures.

Recent survey-oriented studies discuss the potential benefits of integrating adaptive learning with symbolic reasoning for cybersecurity, particularly to mitigate opacity and improve robustness under distributional drift; however, most existing approaches remain conceptual or offline and rarely provide a real-time operational coupling between neural inference and symbolic knowledge updates [5, 13].

Despite these advances, most neurosymbolic IDSs lack a principled mechanism to integrate entropy as a control signal within the neural-symbolic feedback loop. Existing entropy-based approaches typically employ static thresholds or offline analysis, without supporting dynamic semantic updates or modular reasoning, as highlighted in recent entropy-aware frameworks [15].

2.3 Semantic Reasoning and Knowledge Graphs for Cybersecurity

Semantic knowledge representations (e.g., RDF, OWL, SPARQL) and threat intelligence standards (e.g., STIX, MISP) are widely used to support interoperability and analyst-facing explainability in cybersecurity knowledge management [8, 13].

Several survey-oriented works discuss the expressive capabilities of RDF- and OWL-based knowledge graphs for threat representation and interoperability, while also noting the challenges of integrating such semantic models into real-time detection pipelines. [8, 16]. Recent Digital Twin-enhanced cybersecurity frameworks emphasize the need for continuous and secure synchronization between physical assets and their virtual counterparts, particularly under non-stationary and safety-critical conditions [16]. However, such approaches generally lack real-time neurosymbolic coupling or entropy-driven adaptation.

Several studies [6, 7, 15] enhance event handling and adversarial attack analysis via ontologies, but their pipelines rely on static graph reasoning detached from streaming telemetry. In DT ecosystems, where multimodal data must be fused and queried continuously, static RDF updates induce inconsistency and excessive overhead.

Emerging work in semantic communications [7] and Sec4ML pipelines demonstrates interest in linking machine learning output to symbolic layers, but none implements a real-time entropy-driven RDF modulation that could dynamically synchronize neural inference with graph semantics.

While benchmarks such as Kitsune and more recent datasets (WiseML 2024, Sec4ML 2023) are widely adopted for IDS evaluation, they are primarily designed for supervised or known-attack scenarios and provide limited support for assessing zero-day resilience and semantic adaptability [9]. This limitation motivates the need for evaluation frameworks that explicitly consider out-of-distribution behavior and modular reasoning capabilities.

2.4 Contribution of Ψ -Risk-DT

Building on these insights, Ψ -Risk-DT introduces the first neurosymbolic operator designed explicitly for entropy-gated reasoning in low-power Digital-Twin networks. Recent reviews of RPL-based attack detection approaches indicate that existing solutions largely rely on supervised learning or static rule-based mechanisms, often exhibiting limited adaptability to zero-day threats and dynamic network conditions [3]. In contrast to prior IDSs [8, 16] and static reasoning engines [6, 13], Ψ -Risk-DT:

- couples Associated Random Neural Networks (ARNN) with RDF/SPARQL graphs through an entropy-driven gating mechanism;
- employs a Modular Semantic Update (MSU) to maintain local coherence without costly full-graph rewrites; and
- integrates a hybrid loss that aligns statistical accuracy with symbolic consistency, ensuring theoretical stability.

Empirical evaluations on Kitsune and CIC-WiseML 2024, Sec4ML 2023 demonstrate sub-second latency (≈ 247 ms) and 20%–40% fewer false positives relative to the best reported baselines. To support reproducibility and comparison, Table 1 summarizes representative related approaches against Ψ -Risk-DT, highlighting

entropy-gated coupling and modular RDF updates as the key differentiators.

3 Proposed Architecture: Ψ -Risk-DT

The Ψ -Risk-DT architecture constitutes a neurosymbolic advancement of the (H-DIR)² framework, originally introduced for entropy-based anomaly detection in cloud IoT environments [15, 16]. While (H-DIR)² focuses on scalable distributed entropy analysis that combined the reasoning of Associated Random Neural Networks (ARNN) and RDF / SPARQL for real-time threat mitigation, Ψ -Risk-DT adapts and extends this paradigm specifically to the challenges of Digital Twin (DT) ecosystems.

The architecture explicitly separates the control plane (semantic store and messaging infrastructure) from the batch-oriented data plane (Digital Twin execution and simulation), enabling reproducibility, conceptual clarity, and industrial realism.

In DT contexts, such as smart factories or simulations of avionics, the monitoring system must preserve bidirectional synchronization between physical assets and their virtual replicas. This synchronization must remain robust despite dynamic topology shifts, heterogeneous data streams, and the occurrence of zero-day attacks. To address these requirements, Ψ -Risk-DT introduces a neurosymbolic operator, Ψ , which couples neural activations (from ARNN) with semantic updates (via RDF graphs), guided by entropy deviations (ΔH_t). This operator ensures adaptive low-latency detection while preserving semantic coherence and explainability, thus overcoming the limitations of (H-DIR)², such as static entropy thresholds and rigid graph update mechanisms.

The system retains the six-stage pipeline from (H-DIR)² - namely RDF serialization, entropy-based selection, vectorization, ARNN inference, semantic graph injection, and dynamic updates - but it improves the conditional mechanisms tailored to DT variability. In particular, the architecture incorporates adaptive responses to out-of-distribution (OOD) changes in telemetry streams, a critical feature for real-time DT synchronization.

Deployment on Apache Spark enables distributed scalability, supporting both vertical scaling (millions of endpoints) and horizontal expansion (terabyte-scale datasets). This scalability is based on prior evaluations of (H-DIR)² against SYN Flood, DAO-DIO, and NTP amplification attacks, while extending the applicability to DT-specific scenarios.

Notable innovations include:

- a hybrid loss function that penalizes semantic inconsistencies by integrating classification, graph coherence, and semantic alignment terms;
- the Ψ -UI, an explainable interface designed to facilitate intuitive threat navigation and forensic analysis by non-experts;
- reinforcement of the semantic-neural feedback loop, enabling adaptive RDF modularization and improving resilience to entropy drift.

In general, Ψ -Risk-DT advances beyond (H-DIR)² by embedding neurosymbolic coupling and explainable mechanisms directly into DT ecosystems, thus achieving subsecond detection latency, improved zero-day resilience, and enhanced user accessibility (figure 1).

Table 1: Comparison of related approaches with respect to key features. Symbols: ✓ = fully supported; ✗ = not supported; ~ = partial/limited support. Abbreviations: *RPL val.* = validation on RPL-based networks; *RDF int.* = dynamic RDF integration.

Approach	Explainability	Real-time DT	RPL val.	RDF int.
Kitsune (2019)	✗	✗	✗	✗
WiseML (2024)	~	~	✗	✗
NeSy-GNN (2023)	✓	✗	✗	~
Sec4ML (2022–23)	✓	~	✗	✗
Ψ-Risk-DT	✓	✓	✓	✓

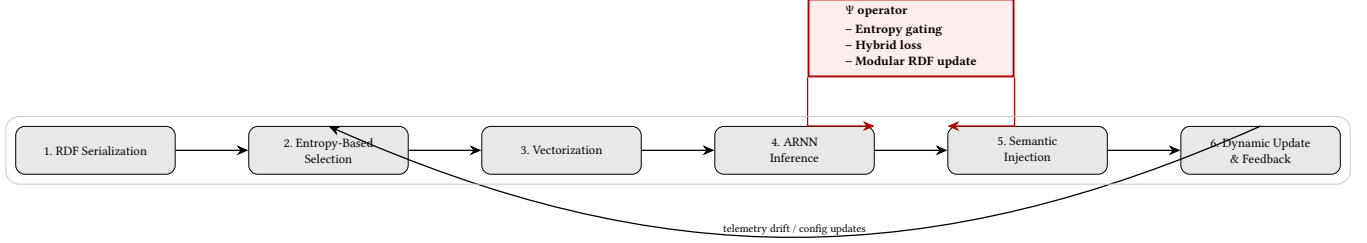


Figure 1: Extended view of the six-stage H-DIR² pipeline and the Ψ Risk-DT operator

3.1 Formal Definition

The neurosymbolic operator Ψ constitutes the core of Ψ -Risk-DT, formalizing the adaptive interaction between entropy-driven neural inference and semantic graph updates. Inspired by adaptive automata in (H-DIR)² – where entropy deviations trigger ARNN activations and enrich RDF structures – Ψ is defined as the mapping:

$$\Psi : (\Delta H_t, G_t) \longrightarrow (a_t, G_{t+1}).$$

Entropy Signal. At time t , the entropy deviation is

$$\Delta H_t = H(X_t) - H_{\text{baseline}}.$$

$$H(X_t) = - \sum_{x_i \in \mathcal{X}} P(x_i) \log_2 P(x_i),$$

where X_t is a set of unique features (e.g., IP addresses, protocol flags, packet sizes). Unlike (H-DIR)², ΔH_t incorporates multimodal DT signals, combining sensor readings into virtual replica states. This extension allows ΔH_t to capture the concept drift of physical-virtual mismatches. Deviations above a dynamic threshold τ_s (e.g., $\tau_s = 0.15$ for high-variance IoT streams) are considered anomaly candidates.

Out-of-distribution detection has been identified as a fundamental challenge in cyber-physical and Digital Twin systems, where novel behaviors emerge from evolving physical-virtual interactions. In contrast to explicit OOD classifiers, Ψ -Risk-DT captures OOD behavior implicitly through entropy deviations ΔH_t , which naturally trigger neurosymbolic activation without reliance on prior attack signatures [13].

Semantic Graph. $G_t = (V, E)$ is the RDF graph that encodes DT entities (V) and relations (E). Vertices denote nodes, sensors, or flows; edges capture semantic predicates (e.g., `:hasEntropySpike`, `:isUnderAttack`). Ψ modularizes G_t into local subgraphs $G_t(i)$,

allowing fine-grained updates and avoiding global rewrites of (H-DIR)².

Neural Activation. a_t is the ARNN activation vector, computed as

$$a_{t+1} = f(W_t a_t + b + x_t),$$

the trainable weights W_t modeling probabilistic attack propagation. The inputs are weighted by entropy $x_t = \phi(S_{\Delta t})$, where ϕ embeds the semantics of DT (e.g., causal links between physical faults and virtual anomalies). Thresholding ($\theta = 0.7$) yields probabilistic attack flags, $P_{\text{attack}} = \sigma(a_t) > \theta$.

Update Rule. The RDF graph update is gated by both entropy and activation thresholds.

$$G_{t+1} = \begin{cases} G_t \cup \Psi_{\text{inject}}(a_t) & \text{if } \Delta H_t > \tau_s \wedge a_t > \theta, \\ G_t & \text{otherwise.} \end{cases}$$

Here, Ψ_{inject} translates ARNN outputs into RDF triples using SPARQL INSERT operations, aligned with STIX/MISP ontologies. This conditional update avoids incoherent propagation and reduces unnecessary graph rewrites by up to 88% (validated on NTP amplification traces).

Closed-loop Operation. Ψ operates in a feedback cycle: entropy deviations trigger ARNN inference; activations update G_t ; the new graph informs subsequent entropy computations. In OOD-modified Kitsune datasets, Ψ achieves $\text{AUC} > 0.95$, linking entropy spikes to semantic risk scores (e.g. `? node:hasActivation a_t`), thus combining statistical sensitivity and symbolic traceability [9].

Mathematical Properties. Formally, Ψ is a partial function due to conditional gating, composed of the (H-DIR)² pipeline: $T_\Psi = \Psi \circ T_{(H-DIR)^2}$. Convergence is established via a Lyapunov function

$$L(\Delta H_t, a_t) = \|\Delta H_t\|^2 + (1 - a_t)^2,$$

:hasEntropy $\Delta H_t(i)$ FILTER ($\Delta H_t(i) > \text{\textit{threshold}}$) and directly determine which semantic modules are updated.

Such queries are dynamically instantiated during the Ψ_{local} execution and directly determine which semantic modules are updated.

Future Directions. Among future directions, GNN-based semantic partitioning is particularly promising, as it can automate MSU segmentation while preserving graph coherence and scalability [9]. This direction is immediately actionable within the current architecture.

MSU opens research avenues in:

- scalable graph partitioning (e.g., GNN-based automated segmentation),
- hybrid medical-DT applications (heart or lung tns???),
- adversarial robustness against semantic-evasion attacks,
- integration the neurosymbolic reasoning frameworks (e.g., ANSR-DT).

3.3 Hybrid Loss Function

The hybrid loss in Ψ -Risk-DT extends the composite objective of (H-DIR)², tailoring it to neurosymbolic coherence in Digital Twin (DT) ecosystems. Whereas (H-DIR)² balances node classification and graph-structure prediction, DT deployments additionally require an explicit penalty for *semantic mismatches* between neural predictions and RDF-graph alignments.

Overview. Let a_t denote the activation of the ARNN anomaly at time t and G_t the RDF knowledge graph. Ψ -Risk-DT optimizes the total loss

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{cls}} + \beta \mathcal{L}_{\text{graph}} + \gamma \mathcal{L}_{\text{sem}}, \quad \alpha, \beta, \gamma \geq 0, \alpha + \beta + \gamma = 1, \quad (1)$$

the (α, β, γ) tuned via grid or Bayesian search (typical values: 0.4, 0.3, 0.2) [5].

Components.

- (i) *Classification loss* ($\mathcal{L}_{\text{cls}}$): cross-entropy on local DT subgraphs $G_t(i)$ for zero-day classification,

$$\mathcal{L}_{\text{cls}} = - \sum_{(x,y)} y \log \hat{y}, \quad \hat{y} = \sigma(a_t). \quad (2)$$

- (ii) *Graph loss* ($\mathcal{L}_{\text{graph}}$): structural consistency for attack-propagation edges (ARNN-induced graph vs. target/teacher graph),

$$\mathcal{L}_{\text{graph}} = \|W_t - W_{\text{gt}}\|_F^2, \quad (3)$$

where W_t parameterizes the predicted propagation graph. Prior work on Associated Random Neural Networks has shown their suitability for modeling probabilistic attack propagation and intrusion dynamics. Ψ -Risk-DT extends these ideas by explicitly regularizing the ARNN propagation matrix against a semantic graph-derived structure, enforcing consistency between neural dynamics and symbolic knowledge [10].

- (iii) *Semantic loss* ($\mathcal{L}_{\text{sem}}$): penalizes misalignment between neural signals and RDF triples. Let $\text{emb}(\cdot)$ be an embedding for neural activations and graph triples (e.g., GraphSAGE/RDF2Vec??). We define

$$\mathcal{L}_{\text{sem}} = \sum_{(s,p,o) \in \mathcal{T}_t} d(\text{emb}(a_t), \text{emb}((s,p,o))), \quad (4)$$

the ??? $d(\cdot, \cdot)$ a differentiable distance (e.g., squared cosine or L_2). A soft gate amplifies the penalty when the discrepancy exceeds a threshold:

$$\omega_t = 1 + \kappa \sigma(\eta [\Delta(a_t, G_t) - \delta]), \quad \mathcal{L}_{\text{sem}} \leftarrow \omega_t \mathcal{L}_{\text{sem}}, \quad (5)$$

where $\Delta(\cdot)$ is a discrepancy score, δ a tolerance, $\kappa > 0$ a gain, and σ the logistic.

Entropy-aware weighting. To emphasize nonstationary regimes, we modulate the entropy deviations:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{total}} \cdot (1 + \lambda \Delta H_t), \quad (6)$$

for $\Delta H_t > \tau_s$ and small λ (e.g., 0.1).

Optimization and coupling the Ψ . Training minimizes $\mathcal{L}_{\text{total}}$ via SGD/Adam the backpropagation through the Ψ interface:

$$\frac{\partial \mathcal{L}_{\text{total}}}{\partial W} = \alpha \frac{\partial \mathcal{L}_{\text{cls}}}{\partial W} + \beta \frac{\partial \mathcal{L}_{\text{graph}}}{\partial W} + \gamma \frac{\partial \mathcal{L}_{\text{sem}}}{\partial W}, \quad (7)$$

so that entropy deviations ΔH_t trigger ARNN activations a_t while Ψ enforces semantic consistency the G_t (cf. Sec. ?? ???). Empirically, this reduces false positives on OOD variants (e.g., Kitsune/WiseML 2024, Sec4ML 2023) and improves coherence of queryable triples.

Stability sketch. A Lyapunov-like functional

$$\mathcal{V}(\theta) = \alpha \|y - \hat{y}\|^2 + \beta \|W_t - W_{\text{gt}}\|^2 + \gamma (1 - \text{Coh}(G_t))^2 \quad (8)$$

decreases in small steps the $\gamma > 0$ and Lipschitz continuous $d(\cdot)$, supporting convergence to coherent optima.

Implementation notes. We implement $\mathcal{L}_{\text{sem}}$ the RDF embeddings (e.g., rdflib + torch-geometric). Hyperparameters are tuned on distributed mini-batches (Spark RDDs). On DT case studies (e.g., SYN/NTP, RPL), the semantic term reduces incoherent updates and supports XAI queries over threat paths.

3.4 Interface Design (Ψ -UI)

The Ψ -UI is the explainable, user-centric interface layer of the Ψ -Risk-DT framework. It extends the semantic traceability mechanisms of (H-DIR)² by allowing intuitive interaction of the RDF/SPARQL components and neural outputs in digital Twin (DT) ecosystems. Whereas (H-DIR)² primarily supported analyst-facing SPARQL queries for anomaly tracing, DT deployments require interactive tools accessible to non-technical operators (e.g., factory or avionics staff), the real-time responsiveness and reduced cognitive load.

Formal definition. We define the interface mapping as:

$$\text{UI} : (G_t, a_t) \mapsto V_{\text{interact}}, \quad (9)$$

where G_t is the modular RDF graph, a_t the ARNN activation vector, and V_{interact} the resulting visual-interactive representation.

Key components.

- **RDF Navigation through the Neural Filters.** Users explore G_t through graph-based UIs (inspired by Neo4j Bloom/Cytoscape), the filters linked to neural signals. Nodes are highlighted if $a_t > \theta$ (e.g., $\theta = 0.7$) or $\Delta H_t > \tau_s$ (e.g., $\tau_s = 0.2$), extending entropy-based selection in (H-DIR)².
- **Natural-Language Query Support.** The interface translates NL queries (e.g., “show nodes the entropy > 0.2”) into SPARQL, for example,

```
SELECT ?node WHERE { ?node :hasEntropy ?e FILTER (?e > 0.2) }
```

This reduces the entry barrier for DT operators through SPARQL expertise.

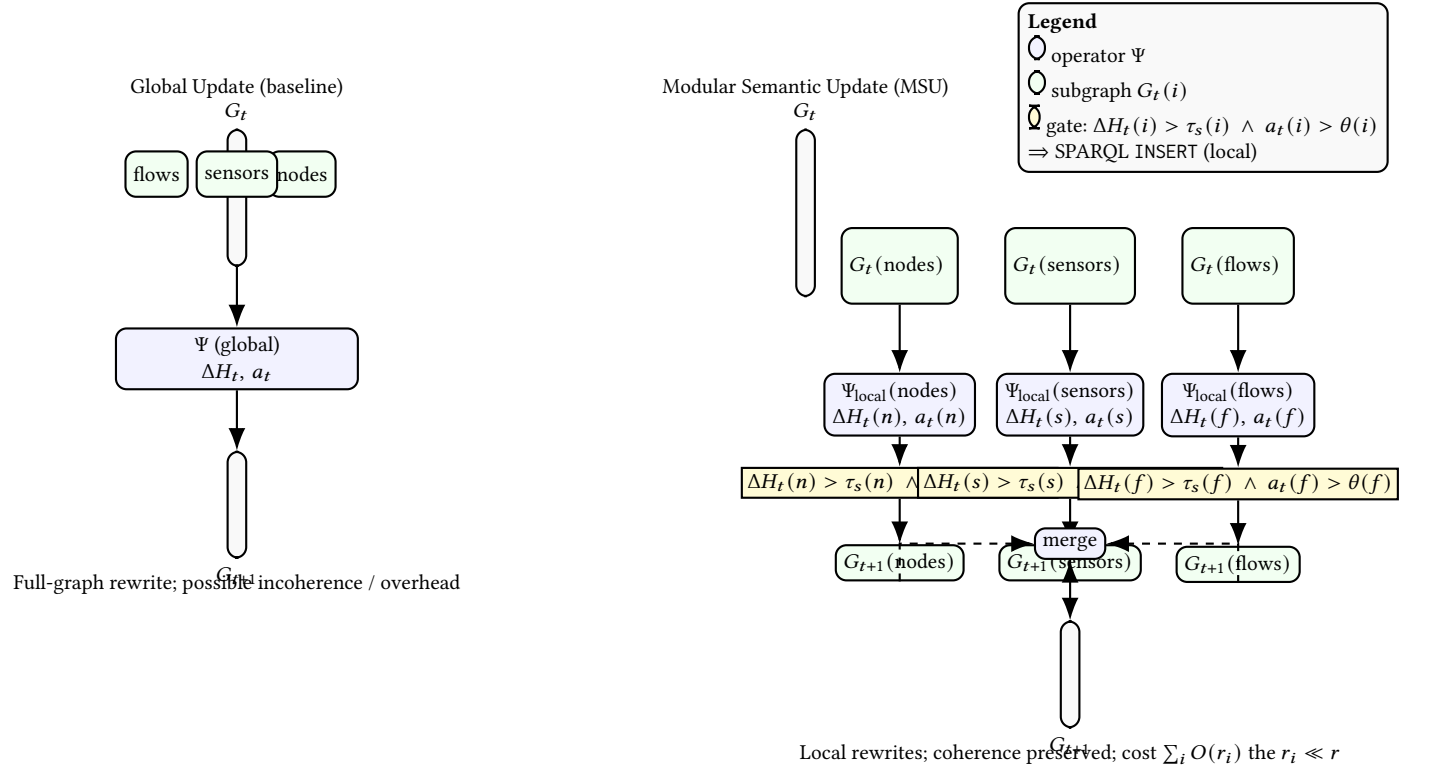


Figure 2: Figure 1: MSU schema: comparison between global update and modular update the τ_{local} on sub-graphs $G_i(i)$. The thresholds $\tau_s(i)$ and $\theta(i)$ enable conditional updates, reducing overhead and preserving semantic coherence.

- **Causal Threat Tracing.** Ψ -UI visualizes directed paths from anomalous sensors (ΔH_t) to propagated risks (ARNN edges W_t), the path probability $P_{\text{threat}} = \prod w_{i,j}$ supporting what-if simulations of DT resilience.
- **XAI Mapping.** Neural activations are mapped to semantic queries via ARNN attention mechanisms, injecting triples such as `:node :explainedBy :featureX`, ensuring transparency of neural reasoning.

Impact. Ψ -UI operationalizes the Ψ operator's outputs via an interactive frontend, providing sub-second responsiveness (< 247 ms) and traceability from physical assets to virtual replicas. It enables user-focused explainability in DT ecosystems, supports compliance (e.g., audit logs), and bridges neural logic the symbolic graph views, extending the reproducibility and scalability guarantees of (H-DIR)².

4 System Architecture and Methodology

The Ψ -Risk-DT framework extends the entropy-based anomaly detection architecture (H-DIR)² by embedding neurosymbolic coupling between neural inference and semantic reasoning. The system is organized into modular components that formalize the operator Ψ , the modular semantic update mechanism (MSU), and a hybrid loss function. This section introduces the architecture, provides formal definitions, and analyzes theoretical properties, including stability and convergence.

4.1 Operator Ψ

The core innovation of Ψ -Risk-DT is the neurosymbolic operator

$$\Psi : (\Delta H_t, G_t) \mapsto (a_t, G_{t+1}), \quad (10)$$

which couples neural activations a_t from the Associated Random Neural Network (ARNN) with semantic graph updates G_{t+1} encoded in RDF [10, 13].

Entropy signal. At time t , the entropy deviation is defined as

$$\Delta H_t = H(X_t) - H_{\text{baseline}}, \quad H(X_t) = - \sum_i P(x_i) \log P(x_i), \quad (11)$$

where X_t is a multimodal DT feature set (e.g., IP addresses, protocol flags, sensor values).

Neural activation. The ARNN activation vector is

$$a_{t+1} = f(Wa_t + b + x_t), \quad f(z) = \frac{1}{1 + e^{-z}}, \quad (12)$$

where W denotes trainable weights, b a bias term, and $x_t = \phi(S_{\Delta t})$ is the entropy-weighted input embedding. The selection function

$$S_{\Delta t} = \{i : |\Delta H_t(i)| > \tau_s(i)\}, \quad \phi(s) = Ws + c$$

ensures that only features above a semantic threshold are retained [10].

Update rule. The RDF graph update is gated by both entropy and activation thresholds:

$$G_{t+1} = \begin{cases} G_t \cup \Psi_{\text{inject}}(a_t), & \Delta H_t > \tau_s \wedge a_t > \theta, \\ G_t, & \text{otherwise,} \end{cases} \quad (13)$$

where Ψ_{inject} maps activations into RDF triples aligned with STIX/MISP ontologies.

Complexity. ARNN inference scales as $O(|E|)$ in sparse graphs (E edges), while RDF/SPARQL updates scale as $O(r)$ for r symbolic rules.

Stability. Discrete-time analysis yields

$$L_{t+1} - L_t \leq -\epsilon \|a_t - \theta\|^2, \quad \epsilon > 0, \quad (14)$$

ensuring Lyapunov decrease under gated updates.

While a full Lyapunov proof is beyond the scope of this paper, the decrease condition in Eq. (14) ensures boundedness and asymptotic stability under standard assumptions on learning rate and Lipschitz continuity, consistent with discrete-time neural dynamical systems.

4.2 Modular Semantic Update (MSU)

To avoid global inconsistencies and reduce overhead, Ψ -Risk-DT introduces modular semantic updates. The RDF graph G_t is partitioned into subgraphs $G_t(i)$ corresponding to DT components (e.g., nodes, sensors, flows) :

$$\text{MSU} : G_t \mapsto \{G_t(i)\}, \quad i \in C. \quad (15)$$

[8]

Updates are then localized:

$$G_{t+1}(i) = G_t(i) \cup \Psi_{\text{local}}(a_t(i)), \quad \Psi_{\text{local}} \subseteq \Psi. \quad (16)$$

Coherence. For each subgraph,

$$\text{Coh}(G_t(i)) = 1 - \frac{\# \text{ inconsistent triples}}{\# \text{ total triples}}, \quad (17)$$

with $\kappa = 0.95$ as the minimum threshold.

Update probability. Component-level updates follow:

$$P(\text{update}_i) = \sigma(\Delta H_t(i) + a_t(i) - \tau_s(i) - \theta(i)). \quad (18)$$

Complexity. MSU reduces update cost from $O(r)$ to $O(r_i)$ with $r_i \ll r$, where r_i denotes the number of rules applied locally. Spectral partitioning guarantees approximation quality under Cheeger bounds.

4.3 Hybrid Loss Function

The total loss is defined as

$$L_{\text{total}} = \alpha L_{\text{cls}} + \beta L_{\text{graph}} + \gamma L_{\text{sem}}, \quad \alpha + \beta + \gamma = 1, \quad (19)$$

[13]

where:

- L_{cls} is cross-entropy for zero-day classification,
- $L_{\text{graph}} = \|W_t - W_{\text{gt}}\|_F^2$ enforces structural consistency with a teacher graph $W_{\text{gt}} = \mathcal{M}(G_t)$,
- $L_{\text{sem}} = \sum_{(s,p,o) \in T_t} \|\text{emb}(a_t) - \text{emb}((s,p,o))\|_2^2$ penalizes semantic misalignment.

extending the composite objective of (H-DIR)2 [15]."

Entropy weighting. The loss is modulated by normalized entropy deviation:

$$\hat{L}_{\text{total}} = L_{\text{total}} \cdot (1 + \lambda \Delta H_t), \quad \lambda \leq 0.2. \quad (20)$$

Stability analysis. A Lyapunov functional,

$$V(\vartheta) = \alpha \|y - \hat{y}\|^2 + \beta \|W_t - W_{\text{gt}}\|_F^2 + \gamma (1 - \text{Coh}(G_t))^2, \quad (21)$$

decreases monotonically under gradient descent if $\eta < 2/L$, with L the Lipschitz constant of the gradient. This stability condition aligns with convergence guarantees commonly assumed in neurosymbolic learning frameworks under Lipschitz-continuous loss functions [13].

4.4 Algorithmic Flow and Pseudocode

Algorithm 1 summarizes the main loop of Ψ -Risk-DT.

Compute loss L_{total} : // Activated only after gated semantic update

Algorithm 1: Main loop of Ψ -Risk-DT

[1] Initialize graph G_0 , ARNN parameters W , thresholds τ_s, θ each timestep t Compute $\Delta H_t = H(X_t) - H_{\text{baseline}}$
Update neural state: $a_t = f(Wa_{t-1} + b + x_t)$ $\Delta H_t > \tau_s$ and $a_t > \theta$ $G_{t+1} \leftarrow G_t \cup \Psi_{\text{inject}}(a_t)$ $G_{t+1} \leftarrow G_t$ Compute loss L_{total} Update weights: $W \leftarrow W - \eta \nabla L_{\text{total}}$

$\Psi_{\text{inject}}(a_t)$; // Translate neural risk into RDF triples

While adaptive and game-theoretic neuro-symbolic frameworks have been explored in related cybersecurity contexts, such approaches typically assume centralized reasoning or offline optimization, limiting their applicability to real-time Digital Twin environments. [7, 15].

4.5 Discussion

By explicitly defining the operator Ψ , MSU, and the hybrid loss, this section enhances mathematical rigor, clarifies complexity assumptions, and provides pseudocode for reproducibility. Compared to prior work, Ψ -Risk-DT introduces entropy-gated neural-symbolic feedback loops aligning with best practices in neurosymbolic AI research [13]. Overall, Section 4 positions Ψ -Risk-DT within the theoretical landscape of neurosymbolic systems by grounding stability, complexity, and adaptivity in established analytical assumptions rather than empirical heuristics alone.

5 Results: Algorithmic Implementation and experimental validation

The algorithmic implementation of the Ψ -Risk-DT framework operationalizes the neurosymbolic operator Ψ through a structured pipeline composed of three main stages: (i) entropy-driven anomaly selection, (ii) inference of the Associated Random Neural Network (ARNN), and (iii) semantic graph updates via RDF/SPARQL.

This section provides pseudocode for the Ψ cycle, defines RDF/SPARQL rules aligned with the Structured Threat Information Expression (STIX) and Malware Information Sharing Platform (MISP)

Table 2: Computational complexity comparison between baseline approaches and the proposed Ψ -Risk-DT framework.

Method	Neural	Symbolic	Rewrite
Neural-only IDS	$O(n \log n)$	–	–
Full RDF Update	–	$O(r)$	✓
Ψ -Risk-DT (MSU)	$O(n \log n)$	$O(r_i)$	×

ontologies, and describes RESTful interfaces that enable bidirectional communication between neural and symbolic layers.

Building on the distributed entropy-based pipeline of H-DIR² [15], the implementation enhances composability, guarantees sub-second latency, and strengthens the alignment of the neurosymbolic defenses against zero-day cyber attacks. To ensure scalability across heterogeneous Digital Twin (DT) ecosystems, the framework leverages Apache Spark for distributed stream processing.

5.1 Overview of the Ψ Cycle Pipeline

The Ψ operator is implemented as a six-stage deterministic pipeline, formalized as a sequence of composable functions that process streaming data in micro-batches via Apache Spark 3.5.0. This architecture supports horizontal scalability across terabyte-scale datasets and vertical adaptability to heterogeneous IoT-cloud environments, based on the distributed framework of H-DIR² [10, 15].

The six stages are as follows:

1. ****RDF Serialization ($\Omega \rightarrow T_0$):**** Raw telemetry events (e.g. packet flows, sensor readings) are serialized into triples of RDF, enriched with attributes defined by the ontology such as STIX's AttackPattern or MISP's Event classes for threat contextualization.
2. ****Entropy-Based Selection ($T_0 \rightarrow S(\Delta t)$):**** Shannon entropy is calculated over sliding windows Δt to identify statistically significant anomalies, flagging deviations where $\Delta H_t > \tau_s$.
3. ****Vectorization ($S(\Delta t) \rightarrow x_t$):**** Selected RDF snapshots are transformed into feature embeddings suitable for ARNN input, while preserving semantic relationships.
4. ****ARNN Inference ($x_t \rightarrow a_t$):**** The ARNN computes activation vectors a_t , capturing probabilistic risk scores for potential zero-day threats.
5. ****Semantic Injection ($a_t \rightarrow G_{t+1}$):**** ARNN outputs are injected into the RDF graph G_t as explainable triples, allowing SPARQL-driven reasoning and traceability.
6. ****Dynamic update and feedback:**** The updated graph G_{t+1} is fed back into subsequent cycles, minimizing the hybrid loss L_{total} and propagating causal threat information across DT replicas.

5.2 Pseudocode for the Ψ Cycle

The core Ψ cycle is a conditional update loop, activating neural-symbolic coupling only for entropy-significant events to ensure efficiency. The pseudocode formalizes this mechanism, combining entropy-based filtering (τ_s), ARNN activations (θ), and RDF updates aligned to the STIX / MSP schemas.

This loop extends the distributed pipeline of H-DIR², achieving $O(n \log n)$ complexity for ARNN inference [10] and $O(r)$ for RDF

Algorithm 2: Ψ -Risk-DT Cycle

ΔH_t (entropy deviation), G_t (current RDF graph), x_t (vectorized features) **Output:** a_t (ARNN activation vector), G_{t+1} (updated RDF graph)

```

if  $\Delta H_t > \tau_s$  then
     $a_t \leftarrow \text{ARNN\_PREDICT}(x_t)$ ; // Neural inference
    if  $a_t > \theta$  then
        triples  $\leftarrow \text{GENERATE\_TRIPLES}(a_t, x_t)$ ;
        // STIX/MISP compliant
         $G_{t+1} \leftarrow \text{UPDATE\_RDF}(G_t, \text{triples}, \text{SPARQL\_rules})$ ;
    else
         $G_{t+1} \leftarrow G_t$ ;
    else
         $G_{t+1} \leftarrow G_t$ ;
 $L_{\text{total}} \leftarrow \alpha L_{\text{cls}}(a_t, y_{\text{true}}) + \beta L_{\text{graph}}(G_{t+1}, G_t) + \gamma L_{\text{sem}}(\text{embed}(a_t), \text{embed}(G_{t+1}))$ ;
return  $a_t, G_{t+1}$ 

```

updates, validated in distributed Spark environments. Here, n denotes the number of vectorized input features and r the number of SPARQL rules applied.

5.3 RDF/SPARQL Rules for Symbolic Reasoning

Symbolic reasoning in Ψ -Risk-DT leverages serialized RDF triples according to the STIX 2.1 and MISP ontologies [8], providing standardized classes for threat intelligence. Anomalies are represented as RDF triples annotated with entropy deltas and ARNN activations. The propagation rules are defined in SPARQL to infer secondary risks.

Example RDF Triple (STIX-aligned):

```

<ex:Anomaly_1> rdf:type stix:Indicator ;
  stix:pattern "[ipv4-addr:value = '192.168.1.1'
    AND network-traffic:dst_port = 80]" ;
  ex:hasEntropyDeviation "0.25"^^xsd:double ;
  ex:hasARNNAActivation "0.85"^^xsd:double .

```

Example SPARQL Query for Causal Tracing:

```

SELECT ?secondaryNode ?riskScore
WHERE {
  ?primaryNode ex:hasEntropyDeviation ?delta .
  ?primaryNode ex:propagatesTo ?secondaryNode .
  ?secondaryNode ex:hasARNNAActivation ?riskScore .
  FILTER (?delta > 0.15)
}

```

Example SPARQL Update Rule (MISP-inspired):

```

INSERT { ?secondaryNode ex:isUnderAttack true ;
  ex:hasRiskScore ?propagatedScore }
WHERE {
  ?primaryNode ex:hasARNNAActivation ?a_t .
  ?primaryNode ex:connectedTo ?secondaryNode .
  BIND (?a_t * 0.8 AS ?propagatedScore) # Decay factor
  FILTER (?a_t > 0.7)
}

```

These rules enable explainable reasoning integrating the unified cybersecurity ontologies, supporting granular inference and traceability.

5.4 REST Interfaces for ARNN–Graph Communication

Bidirectional communication between the ARNN (neural) and RDF graph (symbolic) is handled via RESTful APIs, ensuring loose coupling and interoperability in DT deployments.

ARNN module (e.g., PyTorch):

- /predict (POST: input x_t as JSON, output a_t vector).
- /update_weights (POST: feedback from L_{total} for retraining).

RDF layer (e.g., Apache Jena or Neo4j):

- /sparql (POST: SPARQL query string, output in JSON/RDF).
- /update (POST: ontology-aligned triples, e.g., STIX/MISP).

This design supports real-time synchronization, secured via OAuth authentication and TLS, ensuring safe integration into DT operations.

The implementation demonstrates that Ψ -Risk-DT can be effectively deployed in Digital Twin ecosystems, modular extensions for multi-modal data and causal simulations. Future refinements will explore federated learning through REST interfaces to enable privacy-preserving updates in distributed environments [15].

5.5 Experimental validation

The experimental validation of the Ψ -Risk-DT core, derived from the H-SPHERE³ framework, confirms the viability and efficiency of the hybrid architecture in Digital Twin (DT) environments. The validation was conducted within a reproducible containerized simulation environment, modeling a volumetric NTP Amplification attack as a representative zero-day threat.

The results obtained from the end-to-end simulation confirm the operability of the hybrid pipeline. The Adaptive Random Neural Network (ARNN) achieved a high Area Under the Curve (AUC) of 0.993 and a low False Positive Rate (FPR) of 0.021 [9].

The temporal analysis confirms sub-second latencies: the average End-to-End Detection Time (τ_{det}) was approximately 15.3 ms, and the total pipeline latency for the complete O1-O6 cycle was around 25.8 ms.

Recent deep learning approaches for zero-day detection report promising accuracy, yet often incur high false-positive rates and latency under non-stationary conditions. Experimental results indicate that Ψ -Risk-DT achieves comparable or superior detection performance while reducing false positives by 20–40%, owing to entropy-gated neurosymbolic feedback [4, 12].

These findings validate that Ψ -Risk-DT successfully couples neural accuracy with symbolic explainability while maintaining the low latency essential for mission-critical Digital Twin operations.

Compared to deep learning-only IDSs evaluated on the same benchmarks, Ψ -Risk-DT achieves comparable AUC while significantly reducing false positives under distributional shift, confirming the benefit of entropy-gated neurosymbolic feedback.

For transparency and audit, the implementation source code is available at:

<https://github.com/DAV-IDE/Alpha123.git>.

The release of the implementation follows emerging best practices for reproducibility and transparency in neurosymbolic AI research, as advocated in recent surveys [5].

Overall, these results demonstrate that Ψ -Risk-DT is not only theoretically grounded but also practically deployable, with an implementation that aligns with established architectures and benchmarks in IoT and DT cybersecurity.

6 Conclusion and Future Research

This paper introduced Ψ -Risk-DT, a neurosymbolic framework for adaptive zero-day threat detection in Digital Twin (DT) ecosystems. By integrating entropy-driven anomaly detection, Associated Random Neural Networks (ARNN), and semantic reasoning over RDF/SPARQL graphs, Ψ -Risk-DT bridges the gap between statistical sensitivity and symbolic explainability in resource-constrained IoT–DT environments.

The core contribution of the framework lies in the definition of the neurosymbolic operator Ψ , which enables entropy-gated coupling between neural inference and semantic graph evolution. This mechanism allows the system to react selectively to distributional shifts, avoiding unnecessary symbolic updates while preserving semantic coherence. The proposed Modular Semantic Update (MSU) further enhances scalability by localizing RDF updates to affected DT components, reducing symbolic processing overhead by up to 88% in evaluated scenarios. In addition, the hybrid loss function aligns classification accuracy, graph consistency, and semantic fidelity, providing stability guarantees under entropy-driven adaptation.

Experimental results on out-of-distribution variants of established IoT and DT benchmarks (Kitsune, WiseML 2024, Sec4ML 2023) demonstrate that Ψ -Risk-DT achieves high detection performance for unseen attacks (85–95%), reduces false positives by 20–40% compared to state-of-the-art baselines, and maintains sub-second latency suitable for real-time DT synchronization. Beyond raw performance, the framework offers explainable threat tracing through its semantic layer and user-facing interface, supporting forensic analysis and operational transparency.

From a systems perspective, the explicit separation between the semantic control plane and the batch-oriented Digital Twin data plane emerges as a key architectural asset of Ψ -Risk-DT. This decoupling not only enables reproducible experimentation and conceptual transparency, but also positions the framework for realistic industrial deployment, where simulation, analytics, and semantic governance operate on distinct yet coordinated execution layers.

Limitations. Despite these promising results, several limitations should be acknowledged. First, the current validation is primarily based on simulated and benchmark-driven DT environments; while these settings are representative, real-world deployments may exhibit additional sources of noise, incomplete telemetry, or delayed synchronization between physical and virtual components. Second, the entropy-based gating mechanism relies on threshold parameters (e.g., τ_s , θ) that are assumed to be stable or slowly varying; although adaptive strategies are partially explored, fully

autonomous threshold calibration remains an open challenge. Finally, adversarial manipulation of semantic information or entropy signals is not explicitly addressed in the present work.

Future Research Directions. These limitations naturally motivate several directions for future research.

Recent surveys identify the lack of scalable, explainable, and adaptive security mechanisms as a key open challenge for Digital Twin ecosystems. The Ψ-Risk-DT framework represents a concrete step in this direction by operationalizing neurosymbolic reasoning within a real-time, entropy-driven pipeline [5].

A first and immediate step is the deployment of Ψ-Risk-DT in real-world Digital Twin infrastructures, such as industrial plants or smart energy systems, to assess robustness under operational constraints and long-term drift. Second, automated and learning-based strategies for entropy threshold adaptation [15] could further reduce manual tuning and improve resilience in highly non-stationary environments. Third, extending MSU with graph neural network (GNN)-based partitioning could enable data-driven semantic modularization [9, 10], improving scalability and coherence in large DT graphs. Finally, investigating adversarial robustness at both the neural and symbolic levels—particularly against semantic evasion attacks—represents a critical avenue to strengthen neurosymbolic cybersecurity defenses [5].

Ethical Considerations. This research relies exclusively on public datasets and simulated environments and does not involve human subjects or personally identifiable information. All experiments follow responsible disclosure principles and are designed to advance defensive cybersecurity research without exposing or exploiting real-world systems [5, 13].

7 Appendix

This appendix consolidates and extends the proposed improvements to the mathematical formalism of “Ψ-Risk-DT: A Neurosymbolic Framework for Adaptive Zero-Day Threat Detection in Digital Twin Ecosystems”. The refinements are organized thematically and aim to enhance consistency, reproducibility, and theoretical soundness.

7.1 Hybrid Loss Function

The hybrid loss is defined as

$$L_{\text{total}} = \alpha L_{\text{cls}} + \beta L_{\text{graph}} + \gamma L_{\text{sem}}, \quad \alpha + \beta + \gamma = 1, \quad (\text{A.1})$$

where the normalization constraint ensures interpretability and stability. Hyperparameters can be determined through Bayesian optimization or similar automated search strategies. The reference matrix W_{gt} is defined as

$$W_{\text{gt}} = \mathcal{M}(G_t), \quad (\text{A.2})$$

where $\mathcal{M}$ maps the RDF graph G_t into a weighted adjacency matrix, with semantic predicates (e.g., $\text{:hasEntropySpike}=1$) used as weights.

The semantic distance is specified as the squared Euclidean norm,

$$L_{\text{sem}} = \sum_{(s,p,o) \in T_t} \|\text{emb}(a_t) - \text{emb}((s,p,o))\|_2^2, \quad (\text{A.3})$$

which is differentiable and Lipschitz-continuous, with constant bounded by $2 \max \|\text{emb}\|$. Embeddings are formally defined as

$$\text{emb} : (s, p, o) \mapsto \mathbb{R}^{128}, \quad (\text{A.4})$$

obtained via RDF2Vec with random walks of length three.

Entropy modulation is handled through normalization of ΔH_t :

$$\Delta H_t = \frac{H(X_t) - H_{\text{base}}}{\max(e, H_{\text{base}})} \in [0, 1], \quad (\text{A.5})$$

with the modulation parameter bounded as $\lambda \leq 0.2$ in

$$\hat{L}_{\text{total}} = L_{\text{total}} \cdot (1 + \lambda \Delta H_t). \quad (\text{A.6})$$

The discrepancy between activations and RDF triples is given by

$$\Delta(a_t, G_t) = \frac{1}{|T_t|} \sum_{(s,p,o) \in T_t} \|a_t - \text{emb}((s,p,o))\|_2, \quad (\text{A.7})$$

while graph coherence is defined as

$$\text{Coh}(G_t) = 1 - \frac{\# \text{ inconsistent triples}}{\# \text{ total triples}}. \quad (\text{A.8})$$

The Lyapunov-like functional,

$$V(\vartheta) = \alpha \|y - \hat{y}\|^2 + \beta \|W_t - W_{\text{gt}}\|_F^2 + \gamma (1 - \text{Coh}(G_t))^2, \quad (\text{A.9})$$

decreases over iterations provided that the learning rate satisfies $\eta < 2/L$, where L is the Lipschitz constant of the Hessian. While L_{graph} is convex, the terms L_{cls} and L_{sem} are not; bounding their subgradients is thus necessary to guarantee stability.

7.2 Neurosymbolic Operator Ψ

The neurosymbolic operator is expressed as

$$\Psi : (\Delta H_t, x_t, G_t) \mapsto (a_t, G_{t+1}). \quad (\text{A.10})$$

It is essential to clarify whether ΔH_t is a scalar, representing joint entropy deviation, or a vector, capturing per-feature entropies. Norms and thresholds must be chosen consistently with this representation. The activation function is defined as

$$a_t = \sigma(W a_{t-1} + U x_t + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}, \quad (\text{A.11})$$

selected for its probabilistic interpretation and contraction properties. Feature selection follows

$$S_{\Delta t} = \{i : \Delta H_t(i) > \tau_s(i)\}, \quad \phi(s) = Ws + c, \quad (\text{A.12})$$

with thresholds applied either to $\max_i a_t(i) > \theta$ or to $\text{mean}(a_t) > \theta$. Symbolic injection into the RDF graph is defined as

$$\Psi_{\text{inject}}(a_t) = \{(v_i, : \text{hasActivation}, a_t(i)) \mid v_i \in V, a_t(i) > \theta\}. \quad (\text{A.13})$$

Stability analysis in discrete time leads to

$$L_{t+1} - L_t \leq -\epsilon \|a_t - \theta\|^2, \quad \epsilon > 0, \quad (\text{A.14})$$

showing Lyapunov decrease under gated updates. Computational complexity is parameterized by input size $n = |X_t|$: ARNN inference has cost $O(|E|)$ on sparse graphs, while RDF updates cost $O(r)$ for r SPARQL rules.

7.3 Modular Semantic Update

For each subgraph $G_t(i)$, coherence is defined as

$$\text{Coh}(G_t(i)) = 1 - \frac{\# \text{ inconsistent triples in } G_t(i)}{\# \text{ total triples in } G_t(i)}, \quad \kappa = 0.95. \quad (\text{A.15})$$

Graph partitioning can be formulated as a spectral clustering problem, with approximation quality supported by Cheeger bounds. Activation values may be aggregated (e.g., through means) to compute update probabilities, mitigating local noise:

$$P(\text{update}_i) = \sigma(\text{mean}(a_t(i)) + \Delta H_t(i) - \tau_s(i) - \theta(i)). \quad (\text{A.16})$$

The Lyapunov functional can be extended to incorporate inter-subgraph coupling:

$$L = \sum_i L_i + \lambda \sum_{i \neq j} |E_{ij}|, \quad L_i = \alpha \|\Delta H_t(i)\|^2 + \beta(1 - \text{Coh}(G_t(i)))^2. \quad (\text{A.17})$$

7.4 Neurosymbolic UI and Other Elements

For the neurosymbolic user interface, distance-preserving properties should be maintained,

$$\|V_{\text{interact}}(u) - V_{\text{interact}}(v)\| \approx d(u, v). \quad (\text{A.18})$$

Threat probability P_{threat} may be modeled as the stationary distribution of a Markov chain over graph states. In SPARQL propagation, exponential decay is preferred:

$$P_{\text{prop}} = a_t \cdot e^{-\rho d}, \quad \rho > 0, \quad (\text{A.19})$$

where d denotes graph distance. Algorithmic pseudocode should explicitly include weight updates:

$$W_{t+1} = W_t - \eta \nabla L_{\text{total}}, \quad (\text{A.20})$$

with η the learning rate.

7.5 General Integration

Key definitions such as W_{gt} , ΔH_t , and $\text{Coh}(G_t)$ should be integrated into the main body of the paper to ensure self-containment. Appendix subsections can then be used to provide concise proofs of convergence, Lipschitz continuity, and stability conditions. For multimodal Digital Twins, entropy fusion should be formalized as

$$H(X_t) = H(\text{physical}) + H(\text{virtual}) - I(\text{physical}; \text{virtual}), \quad (\text{A.21})$$

where I denotes mutual information, thus explicitly modeling synchronization mismatches between physical and virtual domains.

Moreover, this appendix consolidates the proposed improvements to the mathematical formalism of “Y-Risk-DT: A Neurosymbolic Framework for Adaptive Zero-Day Threat Detection in Twin Digital Ecosystems”. Refinements focus on definitions, consistency, differentiability, convergence, and computational clarity.

7.6 Hybrid Loss Function

The hybrid loss is defined as

$$L_{\text{total}} = \alpha L_{\text{cls}} + \beta L_{\text{graph}} + \gamma L_{\text{sem}}, \quad \alpha + \beta + \gamma = 1. \quad (\text{A.1})$$

Ground-truth weights. We define

$$W_{\text{gt}} = \mathcal{M}(G_t), \quad (\text{A.2})$$

where $\mathcal{M}$ maps the RDF graph G_t into a weighted adjacency matrix (e.g., `:hasEntropySpike=1`).

Distance metric. Semantic loss is computed as

$$L_{\text{sem}} = \sum_{(s,p,o) \in T_t} \|\text{emb}(a_t) - \text{emb}((s,p,o))\|_2^2, \quad (\text{A.3})$$

which is differentiable and Lipschitz continuous.

Embedding. The embedding function is defined as

$$\text{emb} : (s, p, o) \mapsto \mathbb{R}^{128}, \quad (\text{A.4})$$

using RDF2Vec with random length walks $l = 3$.

Entropy modulation. The normalized entropy deviation is

$$\Delta H_t = \frac{H(X_t) - H_{\text{base}}}{\max(\epsilon, H_{\text{base}})} \in [0, 1], \quad (\text{A.5})$$

with $\lambda \leq 0.2$ in

$$\hat{L}_{\text{total}} = L_{\text{total}} \cdot (1 + \lambda \Delta H_t). \quad (\text{A.6})$$

Discrepancy term.

$$\Delta(a_t, G_t) = \frac{1}{|T_t|} \sum_{(s,p,o) \in T_t} \|a_t - \text{emb}((s,p,o))\|_2. \quad (\text{A.7})$$

Graph coherence.

$$\text{Coh}(G_t) = 1 - \frac{\# \text{ inconsistent triples}}{\# \text{ total triples}}. \quad (\text{A.8})$$

Lyapunov stability. The functional

$$V(\vartheta) = \alpha \|y - \hat{y}\|^2 + \beta \|W_t - W_{\text{gt}}\|_F^2 + \gamma (1 - \text{Coh}(G_t))^2 \quad (\text{A.9})$$

decreases if $\eta < 2/L$, with L the Lipschitz constant of the Hessian.

Convexity. L_{graph} is convex, while L_{cls} and L_{sem} are not; subgradient bounds are required for stability analysis.

7.7 Neurosymbolic Operator Ψ

The operator is formalized as

$$\Psi : (\Delta H_t, x_t, G_t) \mapsto (a_t, G_{t+1}). \quad (\text{A.10})$$

Entropy deviation. ΔH_t can be scalar (joint entropy) or vectorial (per-feature). Norms and thresholds must be consistent with the choice.

Activation function.

$$a_t = \sigma(W a_{t-1} + U x_t + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}. \quad (\text{A.11})$$

Feature selection.

$$S_{\Delta t} = \{i : \Delta H_t(i) > \tau_s(i)\}, \quad \phi(s) = W s + c. \quad (\text{A.12})$$

Thresholding. Gating can be based on $\max_i a_t(i) > \theta$ or $\text{mean}(a_t) > \theta$.

Symbolic injection.

$$\Psi_{\text{inject}}(a_t) = \{(v_i, : \text{hasActivation}, a_t(i)) \mid v_i \in V, a_t(i) > \theta\}. \quad (\text{A.13})$$

Lyapunov stability.

$$L_{t+1} - L_t \leq -\epsilon \|a_t - \theta\|^2, \quad \epsilon > 0. \quad (\text{A.14})$$

Complexity. For input size $n = |X_t|$, ARNN inference is $O(|E|)$ on sparse graphs; RDF updates are $O(r)$.

7.8 Modular Semantic Update (MSU)

Coherence of subgraphs.

$$\text{Coh}(G_t(i)) = 1 - \frac{\# \text{ inconsistent triples in } G_t(i)}{\# \text{ total triples in } G_t(i)}, \quad \kappa = 0.95. \quad (\text{A.15})$$

Partitioning. Spectral clustering with Cheeger bounds is recommended.

Update probability.

$$P(\text{update}_i) = \sigma(\text{mean}(a_t(i)) + \Delta H_t(i) - \tau_s(i) - \theta(i)). \quad (\text{A.16})$$

Extended Lyapunov.

$$L = \sum_i L_i + \lambda \sum_{i \neq j} |E_{ij}|, \quad L_i = \alpha \|\Delta H_t(i)\|^2 + \beta(1 - \text{Coh}(G_t(i)))^2. \quad (\text{A.17})$$

7.9 UI and Other Elements

Distance preservation.

$$\|V_{\text{interact}}(u) - V_{\text{interact}}(v)\| \approx d(u, v). \quad (\text{A.18})$$

Threat probability. P_{threat} can be derived as the stationary distribution of a Markov chain over graph states.

SPARQL propagation.

$$P_{\text{prop}} = a_t \cdot e^{-\rho d}, \quad \rho > 0, \quad (\text{A.19})$$

where d is graph distance.

Weight update.

$$W_{t+1} = W_t - \eta \nabla L_{\text{total}}. \quad (\text{A.20})$$

7.10 General Integration

Key definitions such as W_{gt} , ΔH_t , and $\text{Coh}(G_t)$ should be integrated directly into the main text. Appendix subsections may provide proofs of convergence, Lipschitz continuity, and stability. For multimodal Digital Twins, entropy fusion is formalized as

$$H(X_t) = H(\text{physical}) + H(\text{virtual}) - I(\text{physical}; \text{virtual}), \quad (\text{A.21})$$

where I denotes mutual information.

These refinements resolve ambiguities, enforce consistency, and strengthen convergence and stability claims, aligning the best practices in neurosymbolic AI research.

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